AWS Certified Solutions Architect - Associate

SAA-C03 Master Study Guide

A condensed, exam-focused guide designed to become your primary study material

Prepared from your full-length notes and aligned to the official SAA-C03 domain structure

How to use this guide

Read once end to end, then spend most of your time on the decision tables, scenario triggers, and common traps. The SAA-C03 exam is scenario-based. Focus less on memorizing every service feature and more on recognizing the architecture tradeoff the question is testing.

Official domain	Weight	Study focus
Design Secure Architectures	30%	IAM, least privilege, encryption, private access, S3 protection, network security, multi-account controls
Design Resilient Architectures	26%	Multi-AZ, failover, backups, DR, decoupling, ELB, Auto Scaling, Route 53 health checks
Design High-Performing Architectures	24%	Right compute, storage, database, caching, content delivery, read scaling, streaming
Design Cost-Optimized Architectures	20%	Pricing models, right-sizing, serverless, Spot, lifecycle policies, cost visibility

Master rule

When two answers both work, prefer the one that is more managed, more secure by default, more highly available across AZs, and cheaper without violating stated requirements. If a requirement says lowest latency, strongest compliance, exact ordering, or near-zero RPO/RTO, that requirement overrides cost.

Prepared: June 25, 2026. Verify final exam policies and service scope on the AWS Certification site before test day.

What is inside

Section	Why it matters
1. Exam architecture mindset	The mental model for eliminating distractors.
2. Foundations	Regions, AZs, Well-Architected, HA/FT/DR, shared responsibility.
3. Identity, security, and governance	The largest exam domain: IAM, KMS, S3 security, Organizations, detective and preventive controls.
4. Networking and content delivery	VPC, routes, subnets, endpoints, hybrid connectivity, Route 53, CloudFront, Global Accelerator.
5. Compute, scaling, and deployment	EC2, ELB, ASG, Lambda, containers, Systems Manager, Beanstalk, CloudFormation.
6. Storage and migration	S3, EBS, EFS, FSx, Snow, DataSync, Storage Gateway, AWS Backup.
7. Databases and analytics	RDS, Aurora, DynamoDB, ElastiCache, Redshift, Athena, Glue, DMS, purpose-built databases.
8. Integration and event-driven design	SQS, SNS, EventBridge, Kinesis, Step Functions, API Gateway, Cognito, X-Ray.
9. Monitoring, operations, and cost	CloudWatch, CloudTrail, Config, Trusted Advisor, cost tools, Organizations and Control Tower.
10. Scenario trigger bank and final cram	Fast service selection and common exam traps.

Exam answering process

- Identify the main requirement first: security, resilience, performance, or cost. Ignore details that do not change the architecture.

- Find the constraint words: least operational overhead, no downtime, lowest cost, private, encrypted, global, multi-Region, exact ordering, near real time, one-time migration, ongoing replication.

- Eliminate unmanaged/self-built options when an AWS managed service directly fits the requirement.

- For multi-response questions, each selected answer must be required. Avoid selecting two services that solve the same requirement unless the scenario needs both.

- When cost is asked for, do not choose overbuilt HA or multi-Region unless explicitly required.

The four default architectures to recognize

1) Public web tier + private app tier + private database tier. 2) CloudFront + S3 static site + API Gateway/Lambda + DynamoDB. 3) ALB + Auto Scaling EC2/ECS across multiple AZs + RDS Multi-AZ. 4) Event-driven pipeline: source event -> queue/bus/stream -> compute -> storage/analytics.

1. Exam architecture mindset

Use these defaults to choose between plausible answers.

High-yield design rules

- Secure access with IAM roles, temporary credentials, MFA for humans, least privilege, and explicit network boundaries.

- Design across multiple AZs for high availability. Use multiple Regions only for disaster recovery, global latency, regulatory, or active-active requirements.

- Keep public things public only when they must be public. Databases almost always belong in private subnets without public IPs.

- Decouple producers and consumers with queues, topics, event buses, and streams to absorb spikes and isolate failures.

- Use managed databases and storage instead of self-managed EC2 when the requirement is standard and operational overhead matters.

- Use caching at the right layer: CloudFront at the edge, ElastiCache for app/database reads, DAX for DynamoDB read acceleration, RDS read replicas for relational read scaling.

- Use lifecycle and tiering for storage cost, Savings Plans/Reserved capacity for steady compute, and Spot for interruptible work.

Requirement keywords

Question says...	Think...
least operational overhead	managed/serverless service: Lambda, Fargate, DynamoDB, Aurora Serverless, S3, Athena
private access to AWS service	VPC endpoint. Gateway endpoint for S3/DynamoDB; interface endpoint for most other services.
no single point of failure	Multi-AZ deployment, load balancing, health checks, automated failover
strict ordering or no duplicates	FIFO SQS/SNS; design with message group IDs and deduplication
near real-time streaming	Kinesis Data Streams, MSK, Firehose, Managed Service for Apache Flink
global low-latency HTTP content	CloudFront; add S3 OAC, WAF, ACM in us-east-1 for custom domain
static IPs and TCP/UDP acceleration	AWS Global Accelerator
large offline migration	Snowball Edge or Snowcone; DataSync for online transfer
RTO/RPO very low across Regions	Aurora Global Database, DynamoDB Global Tables, S3 CRR, warm standby or active-active
audit API activity	CloudTrail. Use trails to store beyond event history.
configuration compliance or drift	AWS Config rules and remediation
find sensitive data in S3	Amazon Macie

2. Foundations

Global infrastructure, responsibility boundaries, and resilience language.

AWS global infrastructure

Concept	What to remember for the exam
Region	A separate geographic area that contains multiple AZs. Choose Regions for latency, compliance, service availability, and cost.
Availability Zone	One or more discrete data centers with independent power, networking, and connectivity. Subnets live in one AZ. AZ names can map differently across accounts; AZ IDs are consistent.
Edge location	Point of presence close to users. Used by services such as CloudFront, Route 53, Shield, and Global Accelerator.
Regional edge cache	Larger cache layer between edge locations and origins for CloudFront.

Shared responsibility model

AWS is responsible for security of the cloud	Customer is responsible for security in the cloud
Physical facilities, global infrastructure, hardware, managed service infrastructure, underlying service software.	Data classification and encryption decisions, IAM and access control, network configuration, guest OS and app patching on IaaS, backups, firewall rules, monitoring and logging choices.

Well-Architected pillars

Pillar	Exam translation
Operational Excellence	Automate, observe, test, run workloads effectively, improve continuously.
Security	Least privilege, traceability, defense in depth, encryption, incident response.
Reliability	Recover from failures, scale horizontally, test recovery, manage change.
Performance Efficiency	Use the right service and instance type; scale with demand; offload to managed services.
Cost Optimization	Pay only for what is needed; right-size; use pricing commitments and lifecycle rules.
Sustainability	Maximize utilization and reduce wasted resources.

Availability and recovery language

Term	Meaning	Exam clue
High availability	Minimizes downtime through redundancy and failover.	Use multiple AZs, ELB, ASG, RDS Multi-AZ, Route 53 failover.
Fault tolerance	Continues operating even when components fail.	No interruption or near-zero interruption; more expensive.
RTO	Maximum acceptable time to restore service.	Lower RTO = more pre-provisioned capacity.
RPO	Maximum acceptable amount of data loss measured in time.	Lower RPO = more frequent or continuous replication.
Backup and restore	Lowest cost DR, highest RTO/RPO.	Good when recovery can take hours.
Pilot light	Core components already replicated, scale up during DR.	Moderate cost and RTO.
Warm standby	Scaled-down full environment running.	Faster recovery than pilot light.
Active-active	Multiple Regions serve traffic simultaneously.	Lowest RTO/RPO and highest cost/complexity.

3. Identity, security, and governance

This is the largest exam domain. Know who can do what, from where, and with which encryption.

IAM essentials

Item	Use it for	Exam notes
Root user	Initial account owner and break-glass access.	Enable MFA immediately. Do not use for daily tasks. Do not create access keys.
IAM user	Long-term identity for a person or service account.	Prefer IAM Identity Center for workforce access and roles for applications.
IAM group	Permission management for users.	Groups are collections of users; groups cannot be nested.
IAM role	Temporary credentials for services, apps, federation, or cross-account access.	Best answer when an app, EC2, Lambda, ECS task, or third party needs AWS permissions.
Policy	JSON permission document.	Identity-based policies attach to users/groups/roles; resource-based policies attach to resources such as S3 buckets or KMS keys.
STS	Short-term temporary credentials.	Used by role assumption, SAML/OIDC federation, cross-account access.
Instance profile	Passes an IAM role to EC2.	The EC2 instance receives temporary role credentials through metadata.

Trust policy vs permissions policy

A role needs both. The trust policy says who can assume the role. The permissions policy says what the role can do after it is assumed. If either side is wrong, the role will not work.

Policy evaluation and guardrails

Control	What it does	Key exam point
Explicit deny	Overrides any allow.	If any applicable policy denies, access is denied.
SCP	Sets maximum permissions for accounts/OUs in AWS Organizations.	SCPs do not grant permissions. They only limit what can be granted.
Permissions boundary	Sets maximum permissions for an IAM user or role.	Effective permissions are the overlap of identity policy and boundary.
Session policy	Limits permissions for a role session.	Useful for further restricting temporary credentials.
Resource policy	Resource grants access to principals.	Common for S3 buckets, KMS keys, SQS queues, SNS topics, Lambda functions.
Condition keys	Add context such as MFA, source IP, principal org, external ID.	ExternalId helps prevent confused deputy; aws:PrincipalOrgID limits access to an organization.

Workforce and multi-account access

- IAM Identity Center provides centralized single sign-on for users and groups across AWS accounts and business applications.

- It can connect to external identity providers such as Microsoft AD, Okta, and other SAML/OIDC providers.

- AWS Organizations groups accounts into OUs, centralizes billing, and enables SCPs and delegated administration.

- AWS Control Tower builds and governs a multi-account landing zone with preventive controls, detective controls, account factory, log archive, and audit accounts.

- AWS RAM shares supported resources such as Transit Gateways, subnets, resolver rules, and prefix lists with other accounts.

Encryption and secrets

Service or feature	Use it when	Exam notes
AWS KMS	You need managed encryption keys integrated with AWS services.	KMS keys are regional. Key policies are primary access control. Customer managed keys give more control and cost more.
KMS multi-Region keys	Apps need related keys in multiple Regions.	Still regional resources, but share key material and key ID for cross-Region use cases.
CloudHSM	Strict requirement for dedicated HSM and customer-controlled key management.	More control and responsibility than KMS.
Secrets Manager	Store and rotate database credentials, API keys, and secrets.	Built-in rotation uses Lambda. Costs money. Can replicate secrets across Regions.
SSM Parameter Store	Store configuration and simple secrets.	Standard parameters can be free. SecureString uses KMS. No native automatic rotation.
ACM	Request, import, and manage TLS certificates.	Use with ELB, CloudFront, API Gateway. CloudFront certs must be in us-east-1.

S3 security decision table

Need	Use	Key notes
Prevent accidental public exposure	S3 Block Public Access	Can be enabled at account or bucket level; overrides public policies/ACLs.
Grant object access	Bucket policy or IAM policy	Bucket policies are resource-based; IAM policies are identity-based. An explicit deny wins.
Legacy basic grants	ACLs	Avoid unless required. Bucket owner enforced object ownership disables ACL use.
Default encryption	SSE-S3	S3-managed keys, AES-256, default for new buckets.
More key control/audit	SSE-KMS	Requires KMS permissions to read. KMS key must be in same Region as bucket. Bucket keys can reduce KMS request cost.
Customer supplied encryption key	SSE-C	Client supplies key on every request; HTTPS required.
Client-side control	Client-side encryption	Client encrypts before upload and manages encryption process.
Temporary object sharing	Presigned URL	Uses permissions of the creator and expires at configured time.
Write once, read many	S3 Object Lock	Requires versioning. Governance mode can be bypassed by privileged users; compliance mode cannot be bypassed even by root.
MFA for destructive actions	MFA Delete	Requires versioning; protects permanent deletion and versioning suspension.

Security services map

Service	Primary job	Do not confuse with
CloudTrail	Records account activity and API calls.	CloudWatch metrics/logs.
CloudWatch	Metrics, logs, alarms, dashboards, application/container/lambda insights.	CloudTrail API audit history.
AWS Config	Records resource configuration history and evaluates compliance.	SCPs. Config is detective/reactive, not preventive.
Trusted Advisor	Account recommendations for cost, performance, security, fault tolerance, service limits, operational excellence.	Inspector or GuardDuty.
Inspector	Vulnerability scanning for EC2, ECR images, Lambda.	GuardDuty threat detection.
GuardDuty	Managed threat detection using logs and ML.	It is not an intrusion prevention system.
Macie	Discovers sensitive data in S3.	Does not scan all AWS storage services.
Security Hub	Aggregates security findings across services and accounts.	Does not replace the source services.
Audit Manager	Continuously collects evidence and creates audit reports.	Artifact, which provides AWS compliance documents.
Artifact	Download AWS compliance reports and agreements.	Does not audit your environment.

Network security services

Need	Choose	Notes
Allow/deny traffic to EC2 or ENI	Security group	Stateful; allow rules only; attached at resource/ENI level; can reference other security groups.
Subnet-level stateless filtering	Network ACL	Stateless; allow and deny; numbered rules; first match wins; must define inbound and outbound.
Protect HTTP/HTTPS apps from web exploits	AWS WAF	Layer 7; works with CloudFront, ALB, API Gateway, AppSync, Cognito user pools.
DDoS protection	AWS Shield	Standard is automatic/free for edge services; Advanced adds more resources, response team, and cost protection.
Managed stateful firewall in VPC	AWS Network Firewall	Layer 3/4/7 inspection, domain filtering, intrusion detection/prevention, route traffic through endpoints.
Centralized firewall policies	AWS Firewall Manager	Manages WAF, Shield Advanced, security groups, Network Firewall, Route 53 Resolver DNS Firewall across Organizations.
DNS filtering for VPC queries	Route 53 Resolver DNS Firewall	Blocks/alerts on domains for DNS traffic through the Route 53 Resolver.

Security shortcuts that often win

Use IAM roles instead of access keys. Use VPC endpoints instead of NAT/Internet paths for private AWS service access. Use S3 Block Public Access unless public access is explicitly required. Use CloudTrail plus Config for audit and drift visibility. Use WAF for Layer 7 web rules; use Network Firewall for VPC network inspection.

4. Networking and content delivery

VPC questions usually test route tables, public/private boundaries, and service selection.

VPC components

Component	Remember
VPC	Regional, logically isolated network. Choose IPv4 CIDR from private ranges, commonly /16 to /28. Default VPC is public-friendly and not ideal for production security.
Subnet	Lives in one AZ. Public subnet has route to an Internet Gateway. Private subnet has no direct Internet route. AWS reserves 5 IP addresses per subnet.
Route table	Controls where traffic goes. Longest prefix match wins. Each route has a destination and target. Each VPC has a local route.
Internet Gateway	Attached to VPC, gives route target for Internet-routable IPv4/IPv6 traffic. Public subnet route often 0.0.0.0/0 -> IGW.
NAT Gateway	Allows private IPv4 resources to initiate outbound Internet access. Place in public subnet with Elastic IP; deploy per AZ for HA.
Egress-only Internet Gateway	Allows outbound-only IPv6 Internet access; prevents unsolicited inbound IPv6 connections.
DHCP option set	Controls VPC DNS servers, domain names, NTP. One active option set per VPC; create new to modify.

Security group vs NACL

Feature	Security Group	Network ACL
Level	Resource/ENI	Subnet
State	Stateful	Stateless
Rules	Allow only; implicit deny	Allow and deny; numbered; first match wins
Return traffic	Automatically allowed	Must be explicitly allowed, including ephemeral ports
Best use	Primary workload firewall	Broad subnet guardrail or explicit deny

Private access and endpoints

Need	Choose	Notes
Private subnet to S3 or DynamoDB	Gateway VPC endpoint	Free, route table target, no security group, uses managed prefix lists.
Private subnet to most AWS services	Interface VPC endpoint	Uses PrivateLink and ENIs in subnets; attach security groups; hourly/data cost.
Expose your private service to many VPCs/accounts	AWS PrivateLink endpoint service	Provider fronts service with NLB; consumers create interface endpoints.
Private subnets need Internet downloads	NAT Gateway	External services cannot initiate inbound connections to private instances through NAT.

VPC-to-VPC and hybrid connectivity

Requirement	Best service	Critical exam notes
Connect two VPCs simply	VPC Peering	No overlapping CIDRs. No transitive routing. Update route tables and security controls.
Hub connecting many VPCs and on-prem networks	Transit Gateway	Cloud router. Attach VPCs/VPN/DX Gateway. Supports TGW route tables and peering. Can share with RAM.
Encrypted tunnel over Internet to on-prem	Site-to-Site VPN	Uses VGW or TGW on AWS side and customer gateway on customer side. Enable route propagation where needed.
Individual users need VPN access	AWS Client VPN	Managed OpenVPN-compatible client service; useful for remote access.
Multiple branch sites communicate through AWS	VPN CloudHub	Low-cost hub-and-spoke site-to-site VPN; VPC not required for site-to-site communication.
Dedicated private network connection	Direct Connect	Consistent bandwidth and private connectivity. Traffic is private but not encrypted by default.
Connect one DX to VPCs in multiple Regions	Direct Connect Gateway	Global resource for connecting multiple VPCs to on-prem through DX; not for VPC-to-VPC traffic.

Route 53 DNS

Feature	Use
Hosted zone	Container for DNS records. Public hosted zones route Internet traffic; private hosted zones route within VPCs.
A / AAAA	Maps name to IPv4 / IPv6 address.
CNAME	Maps one DNS name to another DNS name. Cannot be used at zone apex.
Alias	AWS-specific record that can point to AWS resources, including zone apex; no extra DNS query charge.
Simple	Basic routing; can return multiple values randomly.
Weighted	Distribute traffic by relative weights; common for canary or gradual migration.
Failover	Active-passive routing using health checks.
Latency	Route to Region with lowest latency.
Geolocation	Route based on user location; configure default.
Geoproximity	Route based on location of resources/users plus optional bias.
Multivalue answer	Return multiple healthy records.

Hybrid DNS

Feature	What it does
Route 53 Resolver	Default recursive resolver in VPC, reachable at VPC+2 address.
Inbound resolver endpoint	On-prem DNS forwards queries into AWS private hosted zones.
Outbound resolver endpoint	VPC workloads forward selected queries to on-prem DNS.
Resolver rules	Define domains to forward and can be shared across accounts with RAM.
Resolver DNS Firewall	Allow, block, or alert on DNS queries from VPCs.

CloudFront vs Global Accelerator

Choose	When	Key notes
CloudFront	HTTP/HTTPS content delivery, caching, static and dynamic web acceleration, S3 origins, edge functions.	Layer 7 CDN. Use OAC for private S3 origins. ACM cert for custom domain must be in us-east-1. Pair with WAF.
Global Accelerator	TCP/UDP acceleration, static anycast IPs, fast regional failover, non-HTTP apps, IP allowlisting.	Does not cache. Uses AWS global network to reach healthy endpoints such as ALB, NLB, EC2, EIP.

CloudFront details

- Origin can be S3 or custom HTTP/S origin such as ALB, EC2, or any reachable HTTP server.

- TTL controls cache lifetime. Invalidation forces CloudFront to fetch fresh content before TTL expires.

- CloudFront Functions are lightweight JavaScript at viewer request/response. Lambda@Edge supports Node.js/Python and can run at viewer or origin request/response events; deploy Lambda@Edge in us-east-1.

- Georestriction can allowlist or blocklist countries.

Network visibility

Need	Use	Notes
IP traffic metadata	VPC Flow Logs	Capture ACCEPT/REJECT records for VPC, subnet, or ENI; send to CloudWatch Logs, S3, or Firehose; no performance impact.
Packet copies for inspection	VPC Traffic Mirroring	Mirror ENI traffic to a security/monitoring appliance through target and filters.
Query flow logs cheaply	S3 + Athena	Good for long-term analysis and investigations.

5. Compute, scaling, and deployment

Pick the compute model that satisfies operations, scale, latency, and cost.

Compute service chooser

Use case	Choose	Why
Full control over OS, network, and runtime	EC2	IaaS; you manage instance lifecycle, OS patching, and scaling design.
Short event-driven code, no server management	Lambda	Serverless; max 15 minutes per invocation; scales automatically.
Containers with AWS-managed control plane	ECS	Simpler AWS-native container orchestrator; EC2 or Fargate launch type.
Kubernetes requirement	EKS	Managed Kubernetes control plane; nodes can be managed node groups, self-managed, or Fargate.
Run containers without managing servers	Fargate	Works with ECS/EKS; pay for task/pod resources.
Deploy web app from source/image with minimal setup	App Runner	Fully managed web app service.
PaaS for app migration with managed EC2/ELB/ASG	Elastic Beanstalk	Upload code; Beanstalk provisions environment.
Batch jobs and queues	AWS Batch	Runs jobs on Fargate or EC2 compute environments.

EC2 fundamentals

Topic	Exam points
AMI	Template for instance. Public, private, or shared. Prebaked AMIs reduce boot time and ASG cooldown.
User data	Runs bootstrapping scripts at launch. Useful for installing packages, configuring apps, joining domains.
Key pair	Public key stored on instance; private key used for SSH or decrypting Windows admin password.
IMDSv2	Instance Metadata Service with token requirement. Prefer IMDSv2 over IMDSv1 to reduce metadata exposure.
Hibernate	Saves RAM to encrypted EBS root volume and resumes processes later. Faster than cold start for supported instances.
Session Manager	Preferred secure shell access without SSH/RDP exposure. Requires SSM agent, IAM permissions, and network path to SSM endpoints.
Bastion host	Jump server in public subnet for controlled access to private resources. Often replaced by Session Manager.

Instance families

Family type	Examples	Best for
General purpose	M, T	Balanced CPU, memory, network; T is burstable.
Compute optimized	C	High CPU workloads, web servers, batch processing.
Memory optimized	R, X, Z	Large in-memory datasets, caches, analytics.
Storage optimized	I	High sequential/local storage I/O.
Accelerated	P, G	GPU or hardware acceleration, ML, rendering, video.
HPC optimized	Hpc	High performance computing and tightly coupled workloads.

EC2 storage and networking

Feature	Choose it when	Notes
EBS	Persistent block storage for EC2.	AZ-bound network volume. Can encrypt boot and data volumes. Snapshots are incremental and stored in S3-managed storage.
Instance store	Temporary high-performance local block storage.	Physically attached to host. Persists through reboot but lost on stop, hibernate, or terminate.
ENI	Attach network identity to resources.	AZ-bound virtual network card with private IPs, SGs, optional public/EIP.
Elastic IP	Static public IPv4 address.	Avoid unless static IP is required; prefer DNS when possible.
Enhanced networking	Higher packet rate/lower latency.	ENA for up to high bandwidth; EFA for HPC/ML with OS-bypass on Linux.

Placement groups

Type	Optimizes for	Use case
Cluster	Low latency, high throughput within one AZ	HPC, tightly coupled workloads. Use homogeneous supported instance types.
Spread	Separate underlying hardware	Small number of critical instances that should not share hardware.
Partition	Groups of instances on separate partitions/racks	Distributed systems such as Hadoop, Cassandra, Kafka.

EC2 pricing

Model	Best for	Watch out
On-Demand	Short-term, unpredictable workloads	No commitment, highest flexible cost.
Reserved Instances	Steady EC2/RDS usage for 1 or 3 years	Standard gives more savings; Convertible adds flexibility. Zonal RIs reserve capacity.
Savings Plans	Commit to hourly spend for 1 or 3 years	Compute Savings Plans apply across EC2, Fargate, Lambda; EC2 Instance Savings Plans are more specific.
Spot	Interruptible, flexible workloads	Can be interrupted with 2-minute warning. Use diversified/price-capacity-optimized allocation.
Capacity Reservation	Need guaranteed capacity in a specific AZ	Pay On-Demand whether used or not.
Dedicated Host	Licensing/compliance requiring dedicated physical server	Most control and cost; supports BYOL scenarios.
Dedicated Instance	Dedicated hardware but less host-level control	Some licensing support; less control than dedicated hosts.

Elastic Load Balancing

Load balancer	Layer/protocol	Use it for
ALB	Layer 7 HTTP/HTTPS	Web apps, path/host/header routing, OIDC/Cognito auth, Lambda/ECS targets, microservices.
NLB	Layer 4 TCP/TLS/UDP	Extreme performance, static IP/EIP support, TLS pass-through/offload, non-HTTP protocols.
GWLB	Layer 3 IP packets, GENEVE port 6081	Scale and manage virtual network appliances such as IDS/IPS and deep packet inspection.
CLB	Legacy	Avoid for new designs.

ELB features

- Internet-facing load balancers are public; internal load balancers are private.
- Health checks route traffic only to healthy targets.
- Sticky sessions keep a client on the same target when session state requires it, but stateless apps are preferred.
- Deregistration delay/connection draining lets in-flight requests finish before a target is removed.
- SSL offload terminates TLS at the load balancer; pass-through keeps TLS to the backend; bridging terminates and re-encrypts.

Auto Scaling Groups

Concept	Exam points
Launch template	Defines AMI, instance type, key, SGs, IAM role, user data. Preferred over launch configurations.
Min/desired/max	ASG keeps desired capacity between min and max and replaces unhealthy instances.
Target tracking	Maintain a metric target such as average CPU or request count; common best answer.
Step/simple	Scale by discrete adjustments based on alarms.
Scheduled	Predictable traffic changes such as business hours.
Predictive	Forecasts future capacity from historical load.
Health checks	EC2, ELB, VPC Lattice, EBS, or custom. Grace period allows apps to start.
Lifecycle hooks	Run custom actions when instances launch or terminate.
Maintenance policy	Launch before terminate favors availability; terminate and launch favors cost.

Lambda

Feature	Exam points
Runtime and packaging	Use supported runtime, custom runtime, zip, or container image from ECR.
Execution role	Grants the function permissions to call AWS APIs.
Networking	Default runs in AWS-owned VPC with Internet access. Attach to your VPC only when it must reach private resources; then configure subnets, SG, and outbound path/endpoints.
Timeout	Maximum 900 seconds. Choose Fargate, Batch, ECS, or EC2 for long-running workloads.
Memory and CPU	CPU scales with memory setting. More memory can improve performance and sometimes lower cost.
Concurrency	Reserved concurrency caps/guarantees for a function. Provisioned concurrency reduces cold starts.
Versions and aliases	Versions are immutable snapshots. Aliases point to versions and support traffic shifting/canary deployments.
Layers	Share dependencies or common code across functions.
EFS mount	Use when Lambda needs shared file access beyond /tmp.

Containers

Service	Know this
ECR	Managed container image registry. Supports lifecycle policies, KMS encryption, scan on push, tag immutability, cross-account/cross-Region replication.
ECS EC2 launch type	You manage EC2 capacity and ECS agent. Good for deeper control, long-running workloads, and Spot optimization.
ECS Fargate launch type	Serverless containers. You define task and service; AWS manages infrastructure.
Task role	IAM permissions for the application container.
Task execution role	Permissions for ECS/Fargate agent to pull image, write logs, fetch secrets.
EKS	Managed Kubernetes. Use managed node groups/self-managed nodes/Fargate. Pod Identity maps IAM roles to service accounts.
EKS Anywhere / ECS Anywhere	Manage container workloads on customer-managed infrastructure/on-prem.

Deployment and management

Need	Choose	Notes
Repeatable IaC on AWS	CloudFormation	Templates in YAML/JSON. Resources required; parameters, mappings, conditions, outputs optional.
Preview stack changes	Change sets	Like a plan before applying stack update.
Deploy stacks across accounts/Regions	StackSets	Self-managed or service-managed through Organizations.
Govern approved IaC products	Service Catalog	Central catalog of approved templates/services.
Platform team templates	AWS Proton	Manages infrastructure and deployment tooling for developers.
PaaS app deployment	Elastic Beanstalk	Single-instance for dev; load-balanced for production; rolling/immutable/traffic splitting/blue-green via CNAME swap.
Fleet commands and patching	Systems Manager	Requires managed nodes with SSM agent, IAM role, and connectivity to SSM endpoints.
Secrets/config hierarchy	Parameter Store	String, StringList, SecureString. Use Secrets Manager for secret rotation.

6. Storage and migration

Match access pattern, latency, durability, sharing, and migration path.

S3 essentials

Topic	Exam points
Buckets	Bucket names are globally unique; buckets are regional. S3 is object storage, not block storage and cannot host an OS or database engine directly.
Objects	Object key is full path/name. Max object size is 5 TB. Use multipart upload for large files or unreliable networks.
Consistency	Strong read-after-write and list consistency for PUT/overwrite/delete operations.
Versioning	Bucket-level. Once enabled, you can suspend but not return to never-versioned. Delete creates a delete marker; remove it to restore prior version.
Lifecycle	Transition or expire current and noncurrent versions. Use for automated cost optimization.
Replication	Same-Region or Cross-Region; asynchronous; versioning required on source and destination. Existing objects are not replicated unless configured/batch replicated.
Notifications	S3 events can target SNS, SQS, Lambda, or EventBridge.
Performance	Use multipart upload for large uploads, byte-range fetch for parallel downloads, and prefixes to distribute high request rates.
Static website	Can host public static website endpoint; common production design is CloudFront + private S3 origin with OAC.

S3 storage classes

Class	Use when	Notes
S3 Standard	Frequently accessed data	Multi-AZ durability and low latency.
S3 Express One Zone	Very high-performance single-AZ access	Lowest latency S3 storage class; not for multi-AZ resilience requirements.
S3 Standard-IA	Infrequent access but rapid retrieval	Lower storage cost, retrieval fee, multi-AZ.
S3 One Zone-IA	Recreatable infrequent data	Single AZ; cheaper but less resilient.
S3 Intelligent-Tiering	Unknown or changing access patterns	Automatic tiering with monitoring cost.
Glacier Instant Retrieval	Archive with millisecond access	For rarely accessed data needing immediate retrieval.
Glacier Flexible Retrieval	Archive with minutes-to-hours retrieval	Lower cost than Instant Retrieval.
Glacier Deep Archive	Lowest-cost long-term archive	Retrieval can take many hours, up to 48 hours.

Block and file storage

Service	Storage type	Use it for	Critical notes
EBS gp3/gp2	Block	Boot volumes and general workloads	AZ-bound. gp3 lets you provision IOPS/throughput independent of size.
EBS io1/io2	Block	Latency-sensitive, high IOPS apps	Use when gp3 cannot meet performance. Multi-Attach supported for selected io volumes in same AZ with cluster-aware FS.
EBS st1	Block HDD	Throughput-heavy big data, ETL, logs	Cannot be boot volume.
EBS sc1	Block HDD	Cold, infrequently accessed data	Lowest-cost EBS; cannot be boot volume.
Instance store	Block local	Temporary high-performance scratch	Lost on stop/terminate/hibernate.
EFS	NFS file	Shared Linux file system across AZs	Managed, elastic, multi-AZ regional option; use SGs; more expensive than EBS.
FSx for Windows	SMB/NTFS file	Windows apps and Active Directory integration	Supports DFS, ACLs, Windows-native workloads.
FSx for Lustre	Parallel file	HPC, ML, high-throughput workloads	Can integrate with S3.
FSx for NetApp ONTAP	NFS/SMB/iSCSI/NVMe	Enterprise shared storage migrations	ONTAP features managed on AWS.
FSx for OpenZFS	NFS file	Move ZFS/Linux file workloads	Backups stored in S3; single-AZ or multi-AZ options.

EFS performance and cost

- Performance modes: General Purpose for lower latency; Max I/O for very high concurrency with higher per-operation latency.

- Throughput modes: Elastic by default for automatic scaling, Provisioned for known requirements, Bursting based on stored size.

- Storage classes: Standard, Infrequent Access, Archive, with lifecycle policies for movement.

Migration and hybrid storage

Need	Choose	Notes
Offline transfer of large data	Snowball Edge	Petabyte-scale migration/edge processing; ship device to AWS.
Small portable edge device	Snowcone	Portable data transfer and edge compute; can transfer offline or use DataSync online.
Online NFS/SMB/object transfer	DataSync	Agent-based for on-prem; agentless for AWS-to-AWS; supports S3, EFS, FSx and integrity verification.
Legacy SFTP/FTPS/FTP/AS2 into AWS	AWS Transfer Family	Data lands in S3 or EFS. Supports internal or external identity providers.
On-prem file access backed by S3	S3 File Gateway	NFS/SMB interface to objects in S3.
On-prem block backup/migration	Volume Gateway	iSCSI volumes stored as EBS snapshots in S3-managed storage.
Replace tape backups	Tape Gateway	Virtual Tape Library using iSCSI.
Centralized backups	AWS Backup	Backup plans, vaults, cross-account/cross-Region, PITR for supported services, Vault Lock WORM.

7. Databases and analytics

The exam tests workload fit: relational, key-value, document, graph, ledger, time-series, warehouse, cache.

Database chooser

Workload	Choose	Why
Traditional relational OLTP	RDS	Managed MySQL, PostgreSQL, MariaDB, Oracle, SQL Server. Use Multi-AZ for HA and read replicas for read scale.
Relational with higher performance/availability	Aurora	MySQL/PostgreSQL compatible, shared distributed storage, 6 copies across 3 AZs, up to 15 replicas.
Intermittent relational demand	Aurora Serverless	On-demand auto-scaling relational capacity.
Multi-Region relational DR	Aurora Global Database	One primary Region, secondary read Regions, fast cross-Region replication and promotion.
Serverless key-value/document NoSQL	DynamoDB	Single-digit millisecond latency, on-demand/provisioned capacity, Global Tables for active-active.
DynamoDB read cache	DAX	In-memory cache for DynamoDB read acceleration.
Application cache/session store	ElastiCache	Redis/Valkey/Memcached in-memory cache for low-latency repeated reads.
MongoDB-compatible document DB	DocumentDB	Managed MongoDB-compatible workloads.
Graph relationships	Neptune	Billions of relationships, Gremlin/openCypher/SPARQL.
Cassandra-compatible	Keyspaces	Managed/serverless Cassandra Query Language workloads.
Immutable ledger	QLDB	Centralized cryptographically verifiable transaction log.
Time-series	Timestream	IoT/metrics/time-series ingestion and analytics.
Data warehouse OLAP	Redshift	Petabyte-scale analytics. Spectrum queries data in S3 without loading.
Ad hoc SQL over S3	Athena	Serverless query service; use Glue Data Catalog and columnar formats for performance.

RDS high availability, scaling, and backups

Feature	Purpose	Exam trap
Multi-AZ DB instance	High availability and automatic failover to standby in another AZ.	Standby is not used for read scaling.
Multi-AZ DB cluster	HA with two readable standby DB instances.	More read capability than traditional Multi-AZ instance.
Read replica	Read performance and offloading reads.	Can be same AZ, cross-AZ, or cross-Region. Automatic backups required. Not primarily HA.
Automated backup	Point-in-time restore within retention window.	Restores to a new DB instance, not over existing.
Manual snapshot	Long-term backup kept until deleted.	Can copy/share depending on encryption and permissions.
RDS Proxy	Connection pooling for RDS/Aurora.	Very useful for Lambda connection storms and failover resilience.
RDS Custom	OS/database customization for Oracle/SQL Server.	Use when standard RDS is too locked down.

Aurora essentials

Topic	Know this
Cluster	One or more DB instances plus cluster volume. One writer, optional replicas.
Storage	Automatically grows from 10 GB to large scale; 6 copies across 3 AZs with self-healing.
Endpoints	Cluster endpoint for writer; reader endpoint load balances reads; custom endpoint targets chosen instances; instance endpoint targets one instance.
Replicas	Up to 15 Aurora replicas. They share storage with writer, support reads, and can be automatic failover targets.
Failover priority	Tier 0 is highest, tier 15 lowest. If equal, largest replica is promoted.
Backups	Automated backups always enabled. Manual snapshots for longer retention.
Backtracking	Aurora MySQL feature to rewind without restore, up to 72 hours; causes brief disruption.
Cloning	Fast test/dev clones through copy-on-write behavior.
Machine learning	Can integrate with services such as Bedrock, Comprehend, and SageMaker for predictions from SQL.

DynamoDB essentials

Topic	Know this
Table design	Primary key can be partition key or partition + sort key. Items max 400 KB. Attributes can be scalar, document, or set.
Capacity modes	On-demand for unpredictable/spiky traffic; provisioned for predictable workloads and reserved capacity optimization.
Consistency	Strongly consistent reads cost more than eventually consistent reads. Global tables are eventually consistent across Regions.
Query vs Scan	Query by key values and is preferred. Scan reads every item and is expensive.
Auto Scaling	Adjusts provisioned read/write capacity with target tracking.
Streams	24-hour ordered stream of item-level changes; required for Global Tables and useful for Lambda triggers.
Global Tables	Multi-Region, active-active read/write replication. Use for low-latency global apps and regional resilience.
TTL	Deletes expired items within a few days without consuming write capacity.
Backups	On-demand backups and PITR up to 35 days. Exports require PITR and can write to S3.
Security	Encryption at rest by default, IAM/resource policies, VPC endpoints for private traffic.

Analytics and big data

Service	Use it for	Exam notes
Redshift	Managed data warehouse for OLAP	Provisioned clusters or serverless. Snapshots stored in S3. Multi-AZ supported. Reserved capacity available.
Redshift Spectrum	Query S3 data from Redshift	Requires Redshift; does not load data into cluster.
EMR	Managed Hadoop/Spark big data clusters	Primary, core, and task nodes. Spot often good for task nodes; On-Demand for primary/core reliability.
Glue	Serverless ETL and metadata catalog	Crawlers infer schemas; Data Catalog stores metadata; job bookmarks avoid reprocessing.
Athena	Serverless SQL queries over S3	Great for one-time/ad hoc queries. Use Parquet/ORC for performance and lower scanned data cost.
QuickSight	BI dashboards and visualization	SPICE is in-memory engine; Enterprise adds features like column-level security.
OpenSearch	Search and log analytics	Use for full-text search, dashboards, and analytics. Supports IAM/Cognito/KMS/TLS/PrivateLink.
Lake Formation	Govern, secure, and share S3 data lakes	Uses Glue and supports tag-based access control.

Migration databases

Need	Choose	Notes
Discover migration estate	Application Discovery Service	Collects server usage/config data and integrates with Migration Hub.
Rehost servers to AWS	Application Migration Service	Replicates source servers into AWS for lift-and-shift.
Migrate databases with minimal downtime	DMS	Full load, full load + CDC, or CDC only. Replication instance connects source and target.
Same engine migration	DMS homogeneous	Example: SQL Server to SQL Server. Schema conversion usually not needed.
Different engine migration	DMS + SCT or DMS Schema Conversion	Example: Oracle to Aurora PostgreSQL. Convert schema and objects.

8. Integration and event-driven design

Decoupling is one of the most common SAA-C03 patterns.

Messaging and events chooser

Need	Choose	Why
Buffer work between producer and consumer	SQS	Pull-based queue. Consumers receive, process, then delete messages.
Publish one message to many subscribers	SNS	Push-based pub/sub. Fanout to SQS, Lambda, HTTP endpoints, email, SMS, and more.
Route events from AWS/SaaS/custom apps	EventBridge	Event bus with rules, event patterns, archives, replays, scheduled events.
High-throughput ordered streaming	Kinesis Data Streams	Shards, producers, consumers, replay within retention; enhanced fan-out for high throughput.
Load streaming data to S3/Redshift/OpenSearch/SaaS	Data Firehose	Fully managed near real-time delivery; can transform with Lambda and convert to Parquet/ORC.
Kafka-compatible streaming	MSK	Managed Apache Kafka in VPC; alternative to Kinesis when Kafka ecosystem is required.
Migrate existing message broker apps	Amazon MQ	Managed ActiveMQ/RabbitMQ; use when app expects broker protocols/features.
Coordinate multi-step workflow	Step Functions	State machines with retry, catch, parallel, map, choice, wait, and human approval patterns.

SQS essentials

Topic	Know this
Standard queue	Nearly unlimited throughput, at-least-once delivery, best-effort ordering.
FIFO queue	Strict ordering within message group and deduplication. Use when order matters or duplicates cannot be tolerated.
Visibility timeout	After a consumer receives a message, it is hidden. If not deleted before timeout, it reappears.
Long polling	Reduces empty responses and API cost by waiting for messages.
Dead-letter queue	Target for messages that fail processing after max receives. Use for debugging and redrive.
Delay queue/message timer	Postpone message availability.
Encryption	SSE-SQS or SSE-KMS. Use IAM and queue policies for access control.

SNS essentials

Topic	Know this
Standard topic	High-throughput pub/sub; messages pushed to subscribers.
FIFO topic	Ordering and deduplication; can publish only to SQS FIFO queues.
Fanout	SNS topic sends same message to multiple SQS queues so consumers process independently.
Cross-account/Region	SNS supports cross-account and cross-Region delivery with policies and permissions.

EventBridge and orchestration

Feature	Use
Default event bus	Receives AWS service events in each account.
Custom event bus	Separate bus for application events.
Partner event bus	Receives SaaS partner events.
Rules	Match events by pattern or schedule and route to targets.
Archive and replay	Store events and replay them to recover/test event-driven systems.
Schema registry	Discover and organize event structures.
Step Functions Standard	Long-running workflows with exactly-once execution semantics for durable orchestration.
Step Functions Express	High-volume, short-duration workflows.

Kinesis family

Service	Use	Exam notes
Kinesis Data Streams	Custom real-time streaming apps	Shard-based. Records up to 1 MB. On-demand or provisioned capacity. Producers use KPL; consumers use KCL.
Enhanced fan-out	Scale consumers needing dedicated throughput	Pushes records to consumers with high throughput and low latency.
Data Firehose	Managed delivery stream	No custom consumer app required. Good for logs/clickstreams into S3, Redshift, OpenSearch.
Managed Service for Apache Flink	Real-time stream processing	Process streaming data using Apache Flink; can transform/enrich/filter streams.
Kinesis Video Streams	Stream and store live video	For devices/cameras and video analytics.

API front doors and auth

Service	Use	Notes
API Gateway REST API	Feature-rich REST APIs	Stages, deployments, caching, throttling, usage plans, API keys, auth options.
API Gateway HTTP API	Simpler, cheaper HTTP APIs	Fewer features than REST API.
API Gateway WebSocket	Real-time bidirectional apps	Integrates with Lambda/HTTP/AWS services.
Edge-optimized endpoint	Global clients	Uses CloudFront edge locations; ACM cert in us-east-1.
Regional endpoint	Same-Region clients or custom CloudFront	ACM cert in same Region.
Private endpoint	Private API inside VPC	Access through interface VPC endpoint.
AppSync	GraphQL API	Combines data from multiple sources.
Cognito user pool	Application user sign-up/sign-in	Returns JWTs. Supports social IdPs and adaptive auth.
Cognito identity pool	Temporary AWS credentials for app users	Maps authenticated/guest users to IAM roles.
X-Ray	Distributed tracing	Trace requests across services for performance and debugging.

9. Monitoring, operations, governance, and cost

Know which service observes, audits, evaluates, remediates, or budgets.

CloudWatch

Feature	Use	Exam notes
Metrics	Numerical time-series data	Most services provide metrics. EC2 basic is 5 minutes; detailed is 1 minute. Custom metrics are possible.
Logs	Central log storage/querying	Log groups contain log streams which contain log events. Set retention. Encrypt with KMS if needed.
Logs subscription filter	Real-time log forwarding	Send matching logs to Lambda, Kinesis Data Streams, Firehose, or OpenSearch.
Logs export	Export to S3	Can take hours before logs are available for export.
Agent	Collect OS/app metrics and logs	Needed for memory, disk, process info from EC2/on-prem.
Alarm	Trigger actions from metric state	States: OK, ALARM, INSUFFICIENT_DATA. Can notify SNS, EC2 actions, ASG actions.
Composite alarm	Combine alarms	Use AND/OR logic to reduce noise.
Dashboards	Visualize metrics	Cross-account/cross-Region views possible.
Insights	Container, Lambda, Application, Contributor Insights	Use for deeper workload visibility and troubleshooting.

Account governance

Service	Use	Important distinction
Organizations	Multi-account management, OUs, consolidated billing, SCPs, delegated admins.	Management account owns org; member account belongs to one org.
SCP	Prevent accounts/OUs from using actions outside guardrail.	Does not grant access. Root user is also limited by SCPs in member accounts.
Control Tower	Set up and govern landing zone.	Uses Organizations, IAM Identity Center, Config, CloudTrail; has preventive and detective controls.
RAM	Share supported resources with accounts/OUs/organization.	Avoid duplicating shared network services when sharing is supported.
CloudTrail	Audit API and console activity.	Event history is 90 days. Create trails for long-term S3/CloudWatch delivery. Org trail centralizes logs.
Config	Configuration history and compliance.	Regional, not free, reactive. Remediation via SSM Automation.

Cost tools

Service	Use
Cost Explorer	Analyze and visualize historical/forecasted spend; view by service, account, tag, usage type; Savings Plans recommendations.
AWS Budgets	Set cost/usage/reservation/savings plan budgets and alerts; budget actions can run automatically or after approval.
Cost and Usage Report / Data Exports	Most detailed cost and usage data, delivered to S3 for analytics.
Cost Anomaly Detection	ML-based detection of unusual spend by service, account, Region, usage type; alerts through SNS/Slack/Chime.
License Manager	Centrally manage software licenses across AWS and on-prem to avoid overuse and support BYOL.
Compute Optimizer	ML recommendations for compute resources such as EC2, ASG, EBS, Lambda, ECS, RDS.

Cost optimization principles

- Use Savings Plans/Reserved Instances for steady workloads after measuring usage.

- Use Spot for stateless, fault-tolerant, interruptible workloads. Diversify instance types/AZs.

- Use Auto Scaling and serverless to avoid idle capacity.

- Move S3 data through lifecycle tiers based on access patterns; use Intelligent-Tiering when patterns are unknown.

- Reduce data transfer costs by keeping chatty components in the same AZ when appropriate, using CloudFront for Internet delivery, and avoiding unnecessary cross-Region replication.

- Use tags and cost allocation tags so reports can map spend to teams, apps, and environments.

10. Scenario trigger bank

Fast recognition table for the final review pass.

Requirement	Likely answer	Exam note
Public static website, low ops	S3 static hosting + CloudFront	For private origin use OAC; public website endpoint requires public read.
Private S3 content through CDN	CloudFront + S3 OAC	Block direct public bucket access.
Public HTTP app across AZs	ALB + ASG/ECS/EKS across multiple AZs	Use health checks and target groups.
Non-HTTP TCP/UDP high performance	NLB	Supports static IP/EIP and Layer 4 traffic.
Scale third-party inspection appliances	GWLB	GENEVE port 6081; network appliance fleets.
Private subnet downloads packages	NAT Gateway	Deploy one per AZ for HA; private route to NAT.
Private access to S3/DynamoDB	Gateway endpoint	Free and route-table based.
Private access to Secrets Manager/CloudWatch/SSM	Interface endpoint	PrivateLink ENI + security group.
Simple two-VPC private connectivity	VPC Peering	No overlapping CIDR, no transitive routing.
Connect many VPCs and on-prem	Transit Gateway	Central hub with TGW route tables.
Dedicated private connection	Direct Connect	Not encrypted by default; combine with VPN if encryption required.
Remote users need VPN	AWS Client VPN	Managed OpenVPN-compatible service.
Global HTTP caching	CloudFront	Layer 7 CDN and cache.
Global static IP and TCP/UDP acceleration	Global Accelerator	No caching; anycast IPs.
Relational HA	RDS Multi-AZ	Standby for failover, not read scaling.
Relational read scaling	Read replicas or Aurora replicas	Each replica has endpoint; can be cross-Region.
Relational intermittent/unpredictable	Aurora Serverless	Auto-scales capacity.
Global relational DR	Aurora Global Database	Fast cross-Region replication and promotion.
Key-value low-latency serverless	DynamoDB	On-demand for unpredictable; provisioned for predictable.
Active-active NoSQL multi-Region	DynamoDB Global Tables	Enable Streams; read/write any replica.
DynamoDB read acceleration	DAX	In-memory cache for DynamoDB.
Database query/result cache	ElastiCache	Redis/Valkey/Memcached.
Shared Linux files across AZs	EFS	NFS, Linux, managed elastic file system.
Windows SMB file share	FSx for Windows	NTFS, AD, SMB.
HPC shared file system	FSx for Lustre	High throughput, S3 integration.
Long-term object archive	S3 Glacier tiers	Instant/Flexible/Deep Archive based on retrieval time.
Online data migration	DataSync	NFS/SMB/S3/EFS/FSx; verifies integrity.
Offline large transfer	Snowball Edge	Ship device to AWS.
Legacy SFTP into S3	Transfer Family	SFTP/FTPS/FTP/AS2 to S3 or EFS.
Async worker queue	SQS	Pull-based; use DLQ for failures.
Fanout to multiple consumers	SNS + SQS	Each consumer gets its own queue.
App event routing and schedules	EventBridge	Rules route events to targets; archives/replays.
Multi-step workflow	Step Functions	Retries, catch, parallel, map, human approval.
Real-time stream processing	Kinesis Data Streams / MSK	Kinesis native; MSK for Kafka requirement.
Deliver streaming logs to S3	Data Firehose	Managed delivery, transform with Lambda.
Serverless API	API Gateway + Lambda	Add Cognito/IAM/Lambda authorizer as needed.
GraphQL API	AppSync	Combine multiple data sources.
App user sign-in	Cognito user pool	Returns JWT tokens.
Temporary AWS creds for app users	Cognito identity pool	Maps to IAM roles.
Trace distributed requests	X-Ray	Service maps and traces.
API audit history	CloudTrail	Create trail for long-term storage.
Metrics/logs/alarms	CloudWatch	Agent needed for OS memory/disk/process.
Compliance drift	AWS Config	Detective and reactive; remediates with SSM.

Requirement	Likely answer	Exam note
Threat detection	GuardDuty	IDS-like, not IPS.
Vulnerability scanning	Inspector	EC2/ECR/Lambda vulnerabilities.
Sensitive S3 data discovery	Macie	PII and sensitive data in S3.
Central security findings	Security Hub	Aggregates findings across services/accounts.
Layer 7 web protection	WAF	Rules for SQLi, XSS, IP, geo, rate-based.
DDoS protection	Shield	Standard automatic; Advanced for enhanced support/cost protection.
Central firewall policy across org	Firewall Manager	Requires AWS Organizations.
Secrets with rotation	Secrets Manager	Lambda-based rotation, RDS/Aurora integration.
Cheap config/secrets	SSM Parameter Store	SecureString with KMS; no native rotation.
TLS cert for CloudFront	ACM in us-east-1	Regional resources need cert in same Region.
Prevent account actions	SCP	Does not grant permissions.
Cross-account app/vendor access	IAM role + trust policy + ExternalId	Prevents confused deputy.

Common exam traps

If you can explain these, you will avoid many distractors.

Trap	Correct mental model
Security groups vs NACLs	SGs are stateful allow-only resource firewalls. NACLs are stateless subnet firewalls with allow/deny rules and ordered evaluation.
Public subnet	A subnet is public only if it has a route to an Internet Gateway and the resource has a public IP/public path.
NAT Gateway	Must be in a public subnet for Internet access. It is for outbound IPv4 from private resources; do not attach security groups to it.
Gateway endpoint	Only S3 and DynamoDB. Most other services need interface endpoints.
VPC Peering	No transitive routing. VPC B cannot reach VPC C through VPC A just because both peer with A.
Direct Connect	Private and predictable, but not encrypted by default.
Route 53 health checks	Endpoint checks need publicly reachable endpoints. Use CloudWatch alarm health checks for private resources.
RDS Multi-AZ	Availability/failover, not read scaling. Read replicas are for read performance.
Aurora replicas	Share storage with the writer and can be automatic failover targets.
DynamoDB Scan	Reads the whole table and is expensive. Query is preferred.
EBS vs EFS	EBS is block storage in one AZ for EC2. EFS is shared NFS file storage across AZs for Linux.
Instance store	Fast but ephemeral; not for durable data.
S3 versioning	After enabling, you can only suspend. Deletes create delete markers.
S3 replication	Asynchronous and requires versioning on both buckets.
CloudFront vs S3 website endpoint	CloudFront with OAC can keep bucket private; S3 website endpoints require public access.
CloudFront/edge certs	ACM certificate for CloudFront and API Gateway edge-optimized custom domains must be in us-east-1.
Lambda VPC	Only attach Lambda to your VPC when it needs private resources. Then plan NAT or VPC endpoints for outbound AWS/Internet access.
Lambda duration	Max 15 minutes. Use Batch/Fargate/ECS/EC2 for long jobs.
SQS Standard	At-least-once delivery and best-effort order. FIFO is required for strict ordering.
SNS vs SQS	SNS pushes to subscribers; SQS stores messages for polling consumers.
Kinesis vs SQS	Kinesis is streaming/replay/order by shard. SQS is a work queue.
CloudTrail vs CloudWatch	CloudTrail is account/API audit. CloudWatch is metrics, logs, alarms, dashboards.
Config vs CloudTrail	Config records resource configuration and compliance over time; CloudTrail records who called what API.
Config	Detective/reactive. It does not prevent actions by itself.
SCP	Limits maximum permissions. It never grants permissions.
KMS	Key policy must allow use. KMS keys are regional unless using multi-Region related keys.
Secrets Manager vs Parameter Store	Secrets Manager is for secrets and rotation. Parameter Store is cheaper/general configuration.
WAF vs Network Firewall	WAF protects Layer 7 HTTP/S at supported front doors. Network Firewall inspects VPC network traffic.
Macie	Only sensitive data discovery for S3, not every storage service.
Cost answers	Do not choose multi-Region, provisioned high IOPS, or dedicated hosts unless the requirement demands it.

Final cram sheet

Read this the morning of the exam.

One-line service picks

Service	Pick when	Service	Pick when
IAM role	temporary credentials for services, federation, cross-account	RDS Read Replica	relational read scaling
IAM Identity Center	workforce SSO across AWS accounts and apps	Aurora	high-performance MySQL/PostgreSQL-compatible relational
SCP	maximum account/OU permissions; does not grant	DynamoDB	serverless key-value/document NoSQL
KMS	managed encryption keys and audit	DAX	DynamoDB read cache
Secrets Manager	secret storage with rotation	ElastiCache	app/database/session cache
ACM	TLS certs for ELB, CloudFront, API Gateway	Redshift	data warehouse
VPC endpoint	private access to AWS services	Athena	SQL over S3
Transit Gateway	many VPCs/on-prem through a hub	Glue	ETL and Data Catalog
Direct Connect	dedicated private network, not encrypted by default	SQS	pull queue
Route 53	DNS, routing policies, health checks	SNS	push pub/sub
CloudFront	HTTP/S CDN and edge cache	EventBridge	event bus and routing
Global Accelerator	static anycast IPs and TCP/UDP acceleration	Kinesis	real-time streaming
ALB	HTTP/S Layer 7 routing	Step Functions	workflow orchestration
NLB	Layer 4, high performance, static IP	API Gateway	managed API front door
ASG	EC2 fleet self-healing and scaling	Cognito	app user auth and temporary AWS creds
Lambda	short event-driven serverless compute	CloudWatch	metrics, logs, alarms
ECS/Fargate	serverless containers without EC2 management	CloudTrail	API activity audit
EKS	Kubernetes	Config	resource config history and compliance
S3	object storage, lifecycle, static sites, data lake	GuardDuty	threat detection
EBS	AZ-bound block storage for EC2	Inspector	vulnerability scanning
EFS	shared Linux NFS file system	Macie	sensitive data in S3
FSx	managed specialty file systems	WAF	Layer 7 web protection
RDS Multi-AZ	relational HA	Shield	DDoS protection

Common ports

Port	Service	Port	Service	Port	Service
22	SSH	80	HTTP	443	HTTPS
21	FTP	23	Telnet	25/587	SMTP
53	DNS	3389	RDP	3306	MySQL
5432	PostgreSQL	1433	SQL Server	6379	Redis
389	LDAP	6081	GENEVE/GWLB		

Last-minute elimination rule

Distractors often work technically but violate one phrase in the question. Match the exact phrase: least cost, least operations, highest availability, lowest latency, private, encrypted, exact order, no downtime, or multi-Region. The right answer is the one that satisfies all required phrases with the fewest unnecessary moving parts.